# Behind the perfect outcome, who Is the Dempsey Center Really Reaching?

Dempsey Center Annual Client Survey Analysis from 2022–2025

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## 1. Introduction

### 1.1 What We Did?

This project analyzes four consecutive years of annual client survey data collected by the Dempsey Center, a nonprofit cancer support organization based in Maine. The dataset spans 2022 through 2025 and includes 1,179 total respondents across 3 service locations - Lewiston, South Portland, and the virtual Dempsey Connects platform. The analysis combines descriptive statistics with three machine learning methods and statistical test to identify patterns in client satisfaction, service utilization, and demographic composition. All code is publicly available on GitHub and can be reproduced by running the Jupyter notebook from the repository.

### 1.2 Why We Did It?

The Dempsey Center provides all services to anyone affected by cancer. This demand consistently exceeds capacity, meaning that every operational decision involves a tradeoff. The Center already reports strong outcomes in its 2024 annual summary reports which has satisfaction rates above 90%, a life improvement rate near 95%, and a Net Promoter Score above the 80-point world-class threshold. So our analysis for this project is not about improving the outcome, but focus on what is driving them, and whether the strong average conceals meaningful variation across client subgroups.

Specifically, we also ask question - Is there a group of clients who are not benefiting from the Center's services? What factors most strongly predict whether a client will be satisfied? And are those factors ones the Center can actually influence?

## 2. Background

Measuring outcomes in nonprofit healthcare is methodologically challenging. Standard clinical metrics like survival rates or treatment response are not applicable. Patient satisfaction surveys are widely used but frequently criticized for measuring experience quality rather than therapeutic benefit. Net Promoter Score, originally

developed in consumer research, has been adopted in healthcare settings as a proxy for patient trust and loyalty — the argument being that clients who would recommend an organization to family members are expressing something more substantive than momentary satisfaction. Whether NPS actually captures clinical benefit, or merely reflects service experience, is a meaningful distinction. This project treats NPS and the Life Improvement Rate as complementary rather than interchangeable metrics for that reason.

Clients who attend more sessions of a therapeutic service generally report greater benefit, though the causal direction is difficult to establish: frequent attendance may produce improvement, or clients who are improving may simply be more motivated to attend. This ambiguity is present in our data as well, and is addressed directly in the Discussion section. What the data can establish is whether frequency and satisfaction are statistically associated.

# 3. Methods and Analysis

## 3.1. Basic Demographics

Women made up the majority of respondents each year, ranging from 75% to 79%. In 2025, 79.1% identified as women and 20.9% as men. Non-binary and prefer-not-to-answer responses stayed below 2% in all years.

The 65–74 age group was the largest across all four surveys. More notably, the 75-or-older group grew from 14.5% in 2022 to 26.2% in 2025 — nearly doubling in four years. Over the same period, the 45–54 group shrank proportionally. Figure 1 shows the full age breakdown by year.

Looking at cancer relationship, survivors were the largest group overall (n=420), followed by current patients (n=340), Other (n=139), care partners (n=110), and family/friends (n=76). Racial composition was approximately 91% White or Caucasian across all years, which limits any subgroup analysis by race.

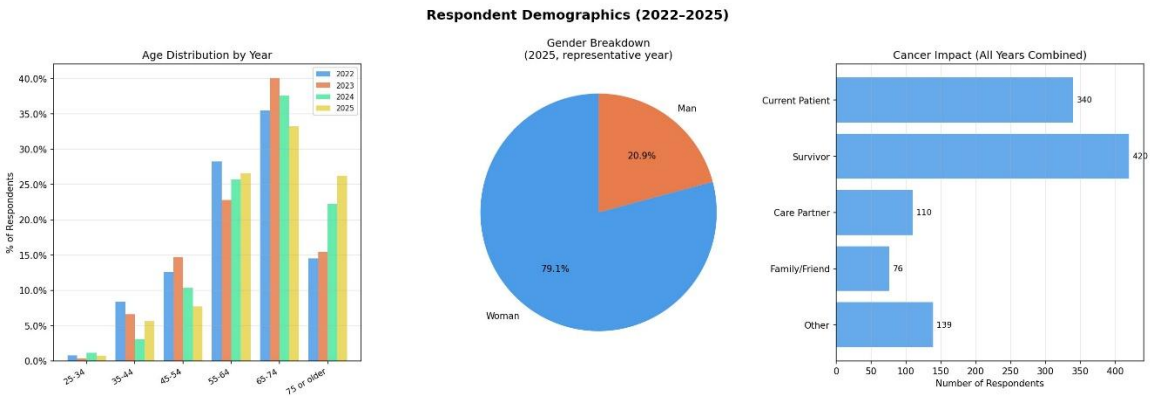


*Figure 1. Respondent Demographics (2022–2025): Age Distribution, Gender Breakdown, and Cancer Impact*

## 3.2. Data Formatting

The raw data is four separate Excel files, one per year. Each file has a two-row header — the first row is full question text, the second is a shorter label. We loaded all files with header=1 in pandas to use the second row as column names.

The biggest formatting headache was that the survey layout changed every year. Questions got added, reworded, or moved around, so the same variable lands at a different column index depending on the year. NPS, for example, is at column 38 in 2022 but column 61 in 2025. The 2025 file also has four extra metadata columns at the front (Respondent ID, Collector ID, Start Date, End Date) that the earlier files don't have, which shifts everything else right by four positions.

To handle this, we built an explicit column index dictionary (COL) that records the correct position for each variable in each year's file. We checked every mapping against the actual column name and spot-checked values before moving on. The full mapping is in the table below:

| Variable | 2022 | 2023 / 2024 / 2025 |
|---|---|---|
| Satisfaction | col 0 | col 0 / col 0 / col 4 |
| Life Better | col 6 | col 1 / col 1 / col 5 |
| NPS | col 38 | col 45 / col 57 / col 61 |
| Visit Frequency | col 27 | col 22 / col 34 / col 38 |
| Age | col 47 | col 54 / col 67 / col 74 |
| Gender | col 49 | col 57 / col 71 / col 78 |
| Cancer Impact | col 43 | col 50 / col 63 / col 68 |

Service usage columns were mapped separately for the 12 services that appear consistently across all four years. These mappings were verified in the same manner. The full mapping is documented in Cell 8 of the Jupyter notebook.

## 3.3. Data Cleaning

Satisfaction text responses were converted to a 1–5 ordinal scale (Very Satisfied = 5 down to Very Unsatisfied = 1), and visit frequency was converted to 1–4 (1 time = 1 up to 10+ times = 4).

The Life Better variable needed extra care. Using a simple == 'Yes' comparison would silently turn NaN values into 0, making missing responses look like "No." We used numpy.where with explicit Yes/No/NaN conditions instead. As it turned out, Life

Better had no missing values at all, but the safeguard is still worth keeping.

Gender wording differed across years like 2022 used "Woman / Female / Feminine" while later years just said "Woman". We unified all labels to consistent strings. The Cancer Impact variable had a similar spacing inconsistency between early and later years, which we standardized as well.

"Prefer not to answer" age entries from 2024 and 2025 were set to NaN. The 77 NPS missing values were excluded from NPS calculations rather than imputed, since filling them in would shift the score distribution. No rows were dropped for missing satisfaction. The final merged dataset has 1,179 rows.

| Variable | Total Missing | % Missing |
|---|---|---|
| Satisfaction | 0 | 0.0% |
| Life Better | 0 | 0.0% |
| NPS | 77 | 6.5% |
| Visit Frequency | 48 | 4.1% |
| Age | 98 | 8.3% |
| Gender | 109 | 9.2% |
| Cancer Impact | 94 | 8.0% |

## 3.4. Tests and Algorithms

### Descriptive Statistics and Trend Analysis

We computed basic summary statistics for all core outcomes by year. Satisfaction rate was the share of respondents scoring 4 or 5. Life Improvement Rate was the share answering "Yes." NPS followed the standard formula: % Promoters (9–10) minus % Detractors (0–6). Charts include year-over-year trend lines, frequency distributions, service utilization rankings, and access barrier comparisons.

### Correlation Analysis

A Pearson correlation matrix was run on 10 variables: Satisfaction, NPS, Frequency, Life Better, and six service-use columns (Massage Therapy, Acupuncture, Individual Counseling, Support Groups, Reiki Session, Movement and Fitness Class). Service use was encoded as binary (1 = used, 0 = not used), and the matrix was computed on all four years combined.

### Principal Component Analysis (PCA)

PCA was applied to a feature matrix built from all four years. The matrix included 12

binary service-use columns plus Satisfaction, NPS, Frequency, and Life Better — 16 features total. All features were standardized with StandardScaler before PCA, and remaining NaN values (mainly in NPS) were filled using most-frequent imputation. We extracted eight principal components and examined the variance explained and feature loadings.

**K-Means Clustering**

K-Means clustering was run on the same standardized feature matrix. We picked the number of clusters using an elbow curve over k=2 through k=8, which suggested k=3. The final model used random_state=42 and n_init=10. We then looked at mean Satisfaction, NPS, Frequency, and Life Better by cluster, and identified the three most-used services in each group.

**Decision Tree Classification**

A Decision Tree classifier was trained to predict whether a client would report high satisfaction (Satisfaction >= 4, labeled 1) or low satisfaction (Satisfaction < 4, labeled 0). Features used were Visit Frequency, Age Group, Cancer Role, and Gender, all encoded as ordinal or nominal integers. The model was trained on all four years of data combined (n=1,098 after dropping rows with missing feature values). An 80/20 train-test split was applied with stratification on the target variable. max_depth=4 was used to limit overfitting. class_weight='balanced' was applied to address class imbalance (89% high satisfaction, 11% low satisfaction).

**Chi-Square Test of Independence**

Two chi-square tests were conducted using scipy.stats.chi2_contingency. Test 1 examined the association between Cancer Role (5 categories) and high/low satisfaction. Test 2 examined the association between Visit Frequency tier (4 categories) and high/low satisfaction. Both tests used all four years of data combined (n=1,179) to ensure adequate cell counts. The significance threshold was $p < 0.05$.

# 4. Results

## 4.1.Core Outcome Metrics

Table 1 summarizes the three core outcome metrics across all four survey years.

| Year | Satisfaction Rate | Life Better Rate |
|---|---|---|
| 2022 | 91.1% | 95.2% |
| 2023 | 85.9% | 95.6% |

| | | |
|---|---|---|
| 2024 | 89.8% | 95.9% |
| 2025 | 90.3% | 95.3% |

| Year | NPS Score | n (NPS valid) |
|---|---|---|
| 2022 | 86 | 269 |
| 2023 | 88 | 275 |
| 2024 | 88 | 266 |
| 2025 | 89 | 292 |

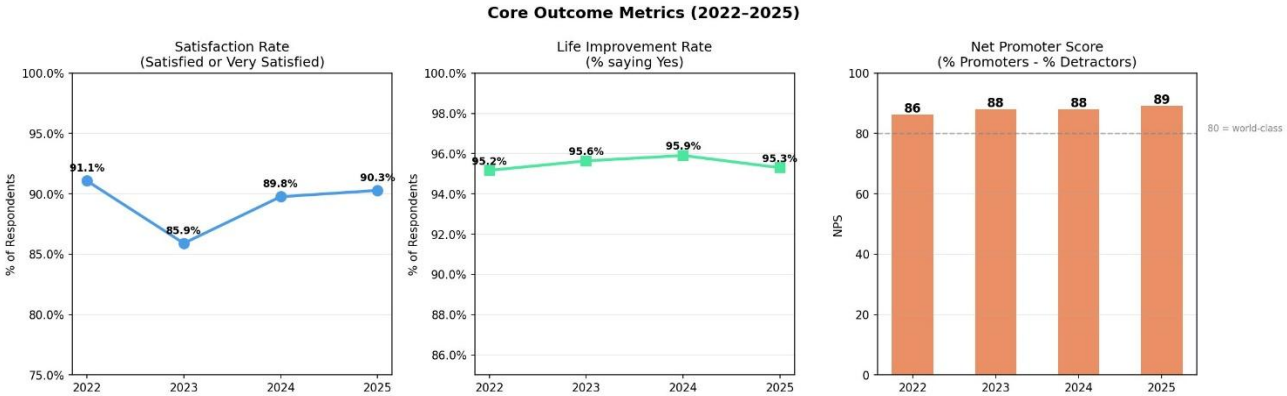


*Figure 2. Core Outcome Metrics (2022–2025): Satisfaction Rate, Life Improvement Rate, and Net Promoter Score*

Satisfaction declined from 91.1% in 2022 to 85.9% in 2023, then recovered to 89.8% in 2024 and 90.3% in 2025. The Life Improvement Rate remained between 95.2% and 95.9% across all years and did not decline in 2023 despite the satisfaction drop. This divergence indicates that the 2023 issue affected service experience without undermining therapeutic outcomes. NPS increased from 86 to 89 over the four-year period. All NPS scores exceeded the 80-point world-class threshold.

## 4.2 Visit Frequency

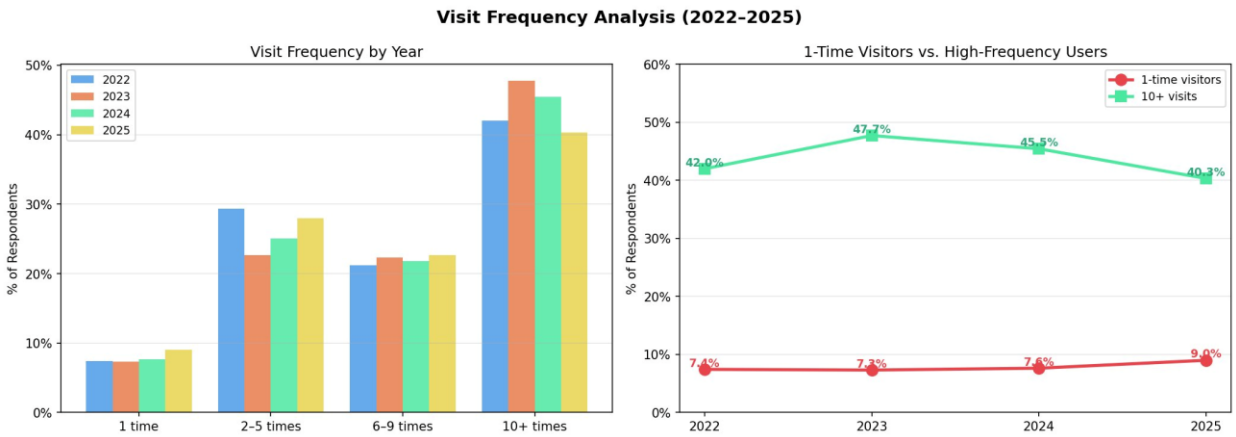

*Figure 3. Visit Frequency Analysis (2022–2025): Distribution by Year and 1-Time vs. 10+ Visit Trend*

The proportion of one-time visitors increased from 7.4% in 2022 to 9.0% in 2025. High-frequency users (10 or more visits) declined from 42.0% in 2022 to a peak of 47.7% in 2023, then fell to 40.3% in 2025. The two trend lines show divergent movement: first-time visitors are increasing while loyal high-frequency users are declining.

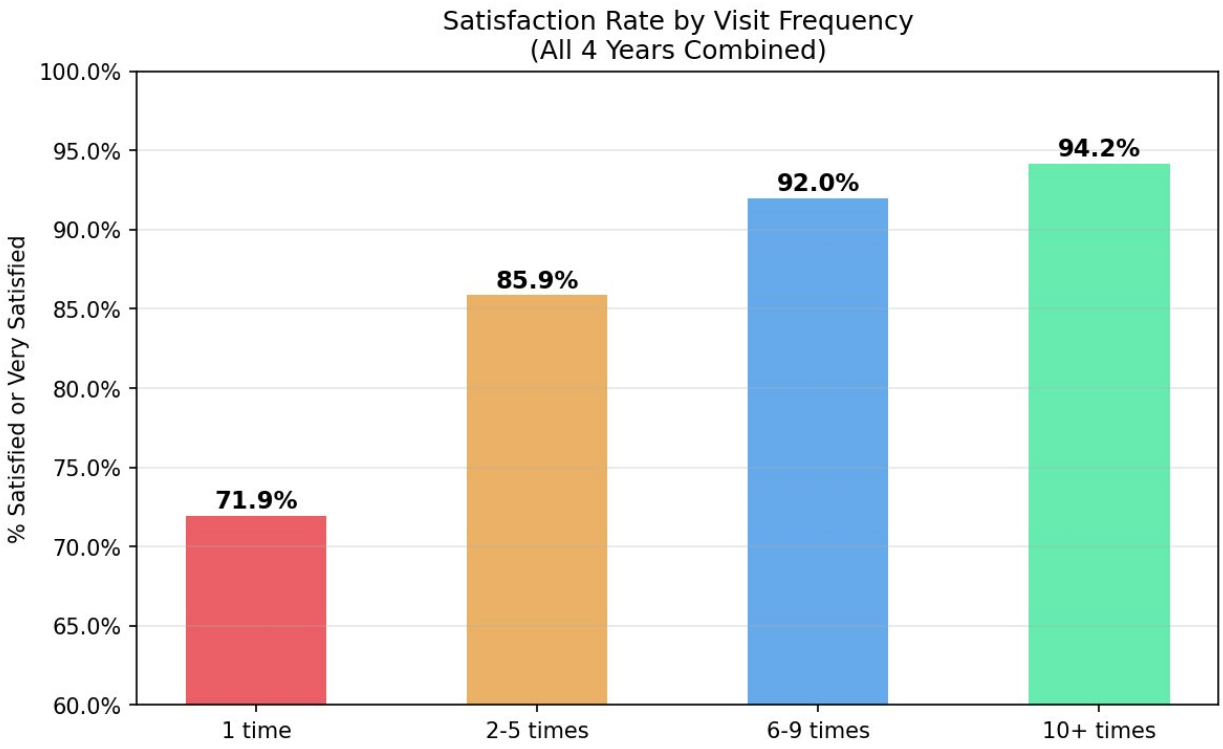


*Figure 4. Satisfaction Rate by Visit Frequency Tier (All 4 Years Combined, n = 1,179)*

Satisfaction rates differ substantially by visit frequency. Clients who visited once reported a 71.9% satisfaction rate. Clients who visited 2-5 times reported 85.9%, those visiting 6-9 times reported 92.0%, and clients with 10 or more visits reported 94.2%. This 22-point gap between one-time visitors and high-frequency users is consistent across all four survey years.

## 4.3. Service Utilization

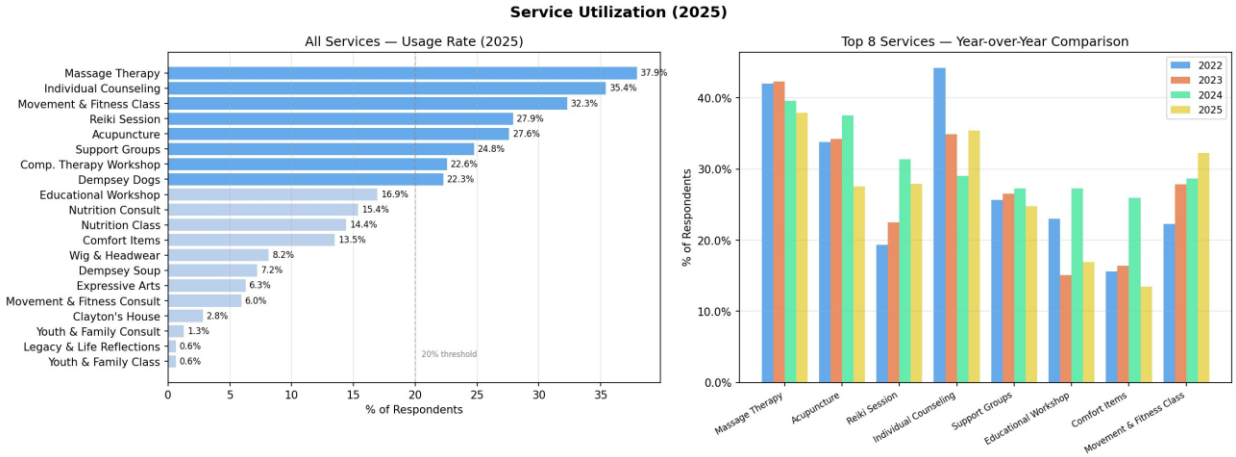


*Figure 5. Service Utilization (2025): All Services Ranked and Top 8 Year-over-Year Comparison*

In 2025, the most used services were Massage Therapy (37.9%), Individual Counseling (35.4%), Movement and Fitness Class (32.3%), Reiki Session (27.9%), Acupuncture (27.6%), and Support Groups (24.8%). Services below the 20% usage threshold are rendered in light blue in Figure 5. Comparing the top 8 services year over year, Individual Counseling usage declined from approximately 44% in 2022 to 35.4% in 2025. Movement and Fitness Class usage grew from 22.3% in 2022 to 32.3% in 2025.

## 4.4. Access Barriers

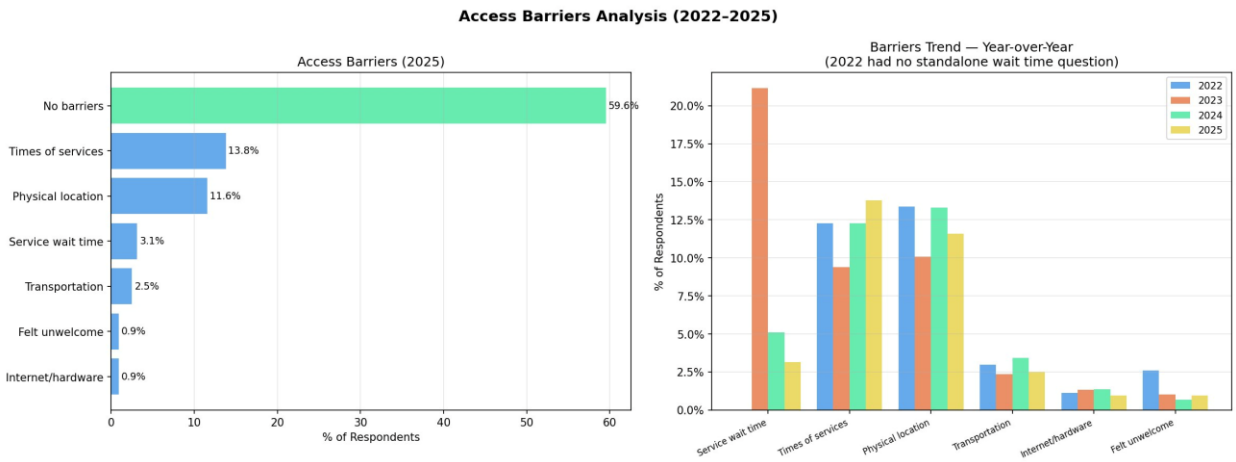


*Figure 6. Access Barriers Analysis (2022–2025): 2025 Breakdown and Year-over-Year Trend*

In 2025, 59.6% of respondents reported no barriers to accessing services. The most frequently cited barriers were Times of services (13.8%), Physical location (11.6%), and Service wait time (3.1%).

The most significant barrier finding is the 2023 spike in Service wait time. This barrier

was not measured as a standalone item in 2022. In 2023, 21.1% of respondents cited it. By 2024 it had fallen to 5.1%, and by 2025 to 3.1%. This spike coincides directly with the 2023 satisfaction decline, providing a likely operational explanation. Transportation as a barrier grew from 3.0% in 2022 to a peak of 13.3% in 2024 before declining to 11.6% in 2025. This growth aligns with the aging of the client population.

## 4.5. Principal Component Analysis

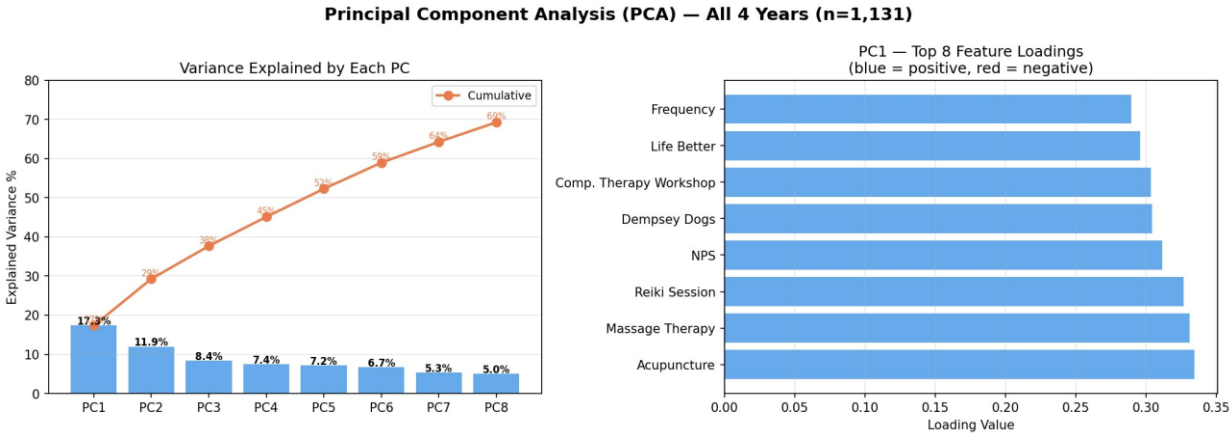


*Figure 7. PCA Results: Explained Variance by Component and Top Feature Loadings for PC1*

PC1 explains 17.3% of total variance and PC2 explains 11.9%. Eight components together explain approximately 69% of variance. This relatively low cumulative explained variance is typical of survey data, which tends to have high noise.

The top features loading on PC1 are Frequency, Life Better, Comp. Therapy Workshop, Dempsey Dogs, NPS, Reiki Session, Massage Therapy, and Acupuncture. All load positively. This suggests PC1 captures overall engagement: clients who score high on PC1 are frequent visitors who use complementary therapies and report strong outcomes. PC1 separates high-engagement clients from low-engagement clients.

## 4.6. Correlation Analysis

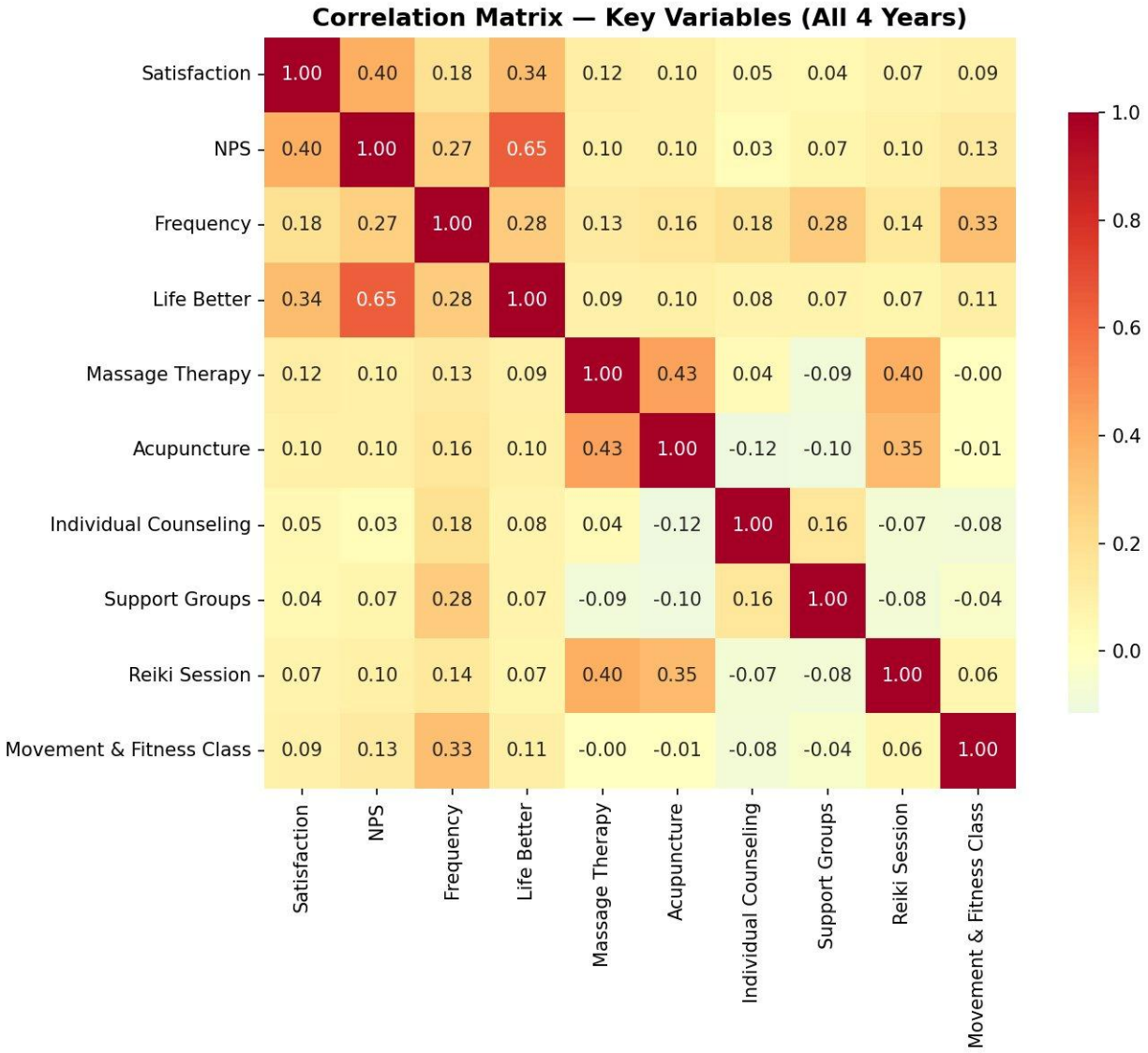


*Figure 8. Correlation Matrix — Key Variables (All 4 Years Combined)*

The strongest correlation in the matrix is between NPS and Life Better (r = 0.65). This indicates that clients who would recommend the Center are largely the same clients whose lives improved, confirming that NPS reflects genuine impact rather than surface satisfaction. Satisfaction and NPS correlate at r = 0.40. Frequency and Movement and Fitness Class correlate at r = 0.33, the strongest service-frequency relationship in the matrix, suggesting fitness classes are the most retention-linked service. Individual Counseling and Satisfaction show near-zero correlation (r = 0.05). This does not imply counseling is ineffective; it reflects that satisfaction is not the appropriate outcome measure for psychosocial services.

## 4.7.K-Means Clustering

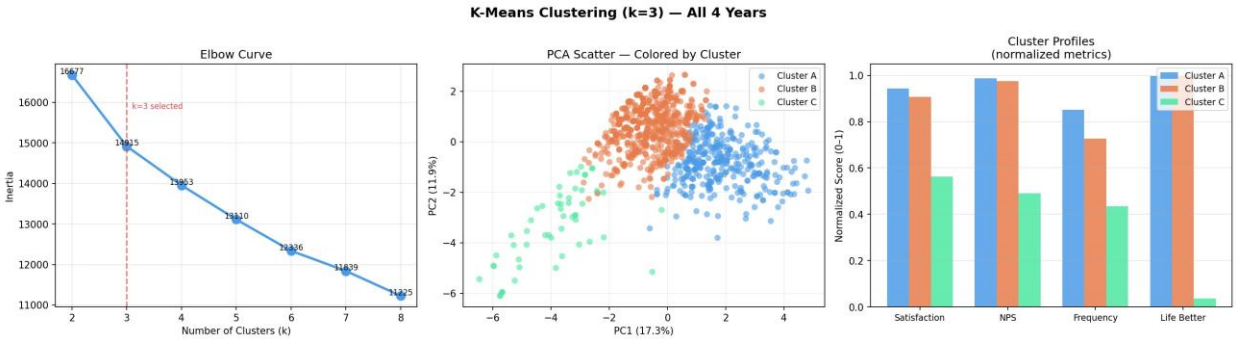


*Figure 9. K-Means Clustering (k=3, All 4 Years): Elbow Curve, PCA Scatter, and Cluster Profiles*

The elbow curve showed a meaningful reduction in inertia from k=2 to k=3, with diminishing returns at k=4 and beyond. k=3 was selected. The PCA scatter plot shows reasonable separation between clusters, with Cluster 2 visibly distinct from Clusters 0 and 1.

| **Metric** | **Cluster 0 (n=365)** | **Cluster 1 (n=760)** | **Cluster 2 (n=54)** |
|---|---|---|---|
| Satisfaction (mean) | 4.72 / 5 | 4.53 / 5 | 2.81 / 5 |
| NPS (mean) | 9.88 | 9.74 | 4.91 |
| Avg Visit Frequency | 3.40 | 2.91 | 1.74 |
| Life Better Rate | 100% | 100% | 4% |
| Top 3 Services | Massage, Acupuncture, Reiki | Counseling, Fitness, Support | Counseling, Massage, Comfort Items |

Cluster 0 (High Engagement) represents loyal users of physical and complementary therapies who visit frequently and report the highest outcomes. Cluster 1 (Moderate Engagement) is the largest group, oriented toward counseling and wellness services, with similarly high outcomes. Cluster 2 (Low Engagement / Low Benefit) is a small but critically important group: only 4% say their life improved, satisfaction averages 2.81, and visit frequency averages 1.74. The appearance of Comfort Items in this cluster's top services suggests these clients may be recently diagnosed patients who have not yet found a service fit.

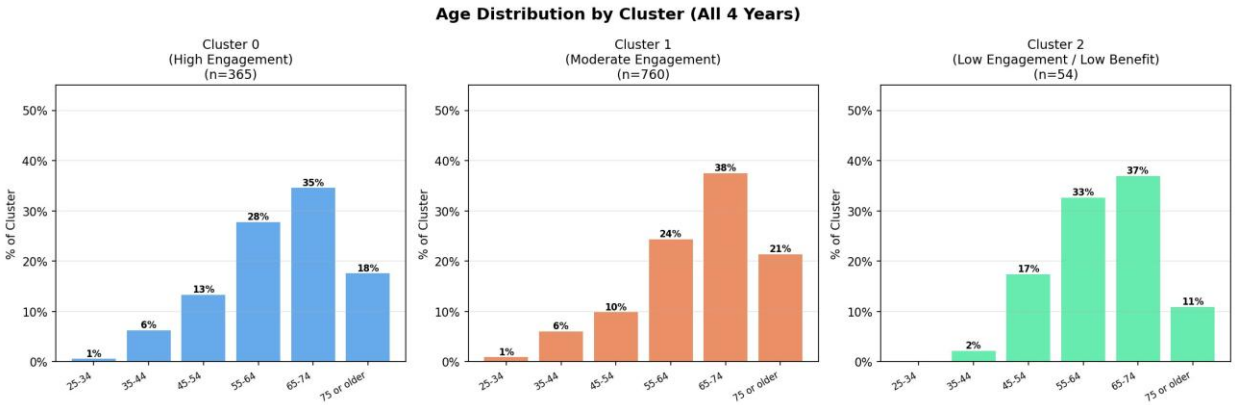


*Figure 10. Age Distribution by Cluster (All 4 Years)*

Cluster 2 has a notably higher proportion of clients aged 45 to 54 (17.4%) compared to Clusters 0 and 1 (approximately 10-13%). The proportion of clients 75 or older is lower in Cluster 2 (10.9%) than in Clusters 0 (17.6%) or 1 (21.4%). This challenges the assumption that older clients are the most difficult to serve; middle-aged clients appear disproportionately represented among low-benefit users.

## 4.8.Decision Tree Classification

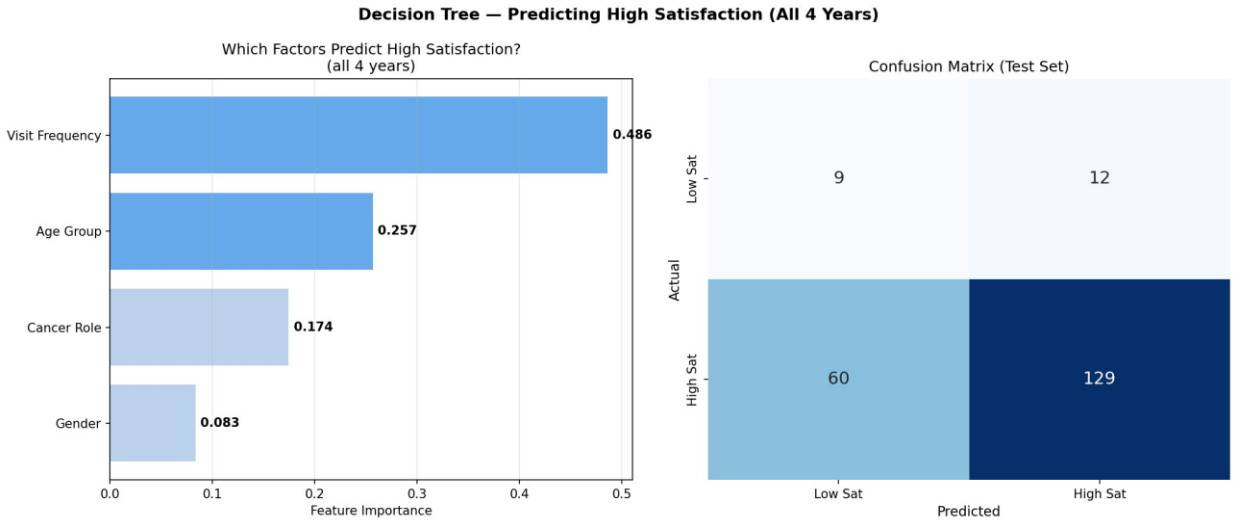


*Figure 11. Decision Tree: Feature Importance Ranking and Confusion Matrix*

The Decision Tree trained on all four years of data (n=1,098 after filtering for complete feature values) achieved 76.9% accuracy on the test set. Feature importance rankings are: Visit Frequency (0.486), Age Group (0.257), Cancer Role (0.174), and Gender (0.083). Visit Frequency has nearly double the importance of the next-highest factor.

The confusion matrix shows that among 21 true low-satisfaction cases in the test set, 9 were correctly identified (recall = 0.43). The class imbalance (89% high satisfaction, 11% low satisfaction) limits the model's ability to identify low-satisfaction clients reliably. Gender had zero predictive importance in the model.

## 4.9.Chi-Square Tests

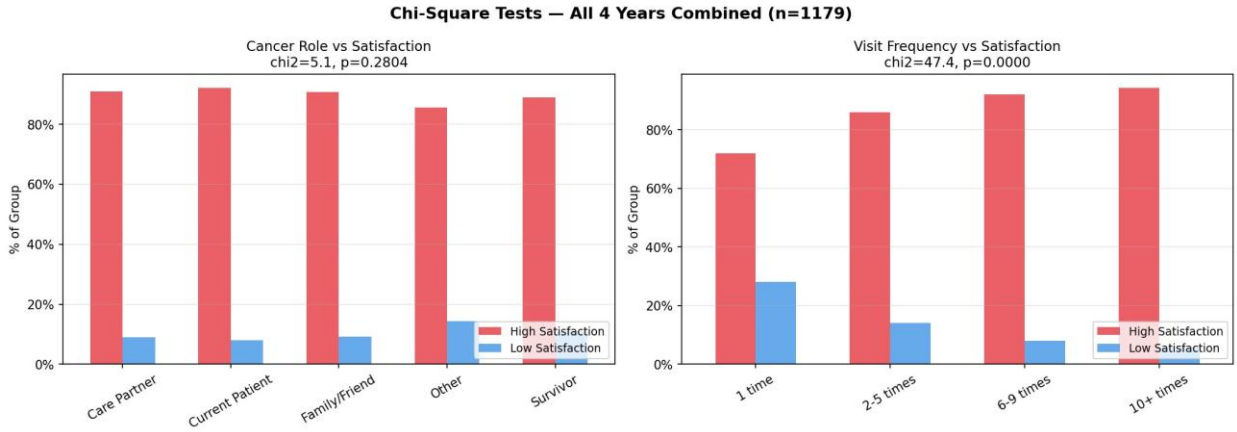


*Figure 12. Chi-Square Tests: Cancer Role vs. Satisfaction and Visit Frequency vs. Satisfaction (n = 1,179)*

| Test | Chi-Square | p-value | Significant? |
|---|---|---|---|
| Cancer Role × Satisfaction | 5.07 | 0.2804 | No ($p > 0.05$) |
| Visit Frequency × Satisfaction | 47.44 | < 0.001 | Yes ($p < 0.05$) |

Cancer Role does not have a statistically significant association with satisfaction ($p = 0.2804$). Despite ranking third in Decision Tree feature importance, this result indicates the observed differences across cancer role groups could plausibly occur by chance. The Center serves current patients, survivors, and care partners at statistically comparable satisfaction levels.

Visit Frequency has a highly significant association with satisfaction ($p < 0.001$). This is the only variable validated by both machine learning (highest feature importance) and statistical testing. The stacked bar chart shows that one-time visitors have a markedly higher Low Satisfaction rate than clients who visit multiple times, with the pattern holding across all frequency tiers.

# 5. Discussion

## 5.1. What the Results Mean

The core metrics tell a story of strong organizational performance, but the interesting findings lie beneath the averages. The overall 95% Life Improvement Rate and 90% Satisfaction Rate are genuine, but they obscure a cluster of 54 clients who report almost no benefit from their visits. This group is not detectable through descriptive statistics alone; it requires the kind of unsupervised segmentation that K-Means provides.

The divergence between satisfaction and life improvement in 2023 is analytically significant. Satisfaction fell by 5.2 percentage points while Life Better remained stable. This pattern is consistent with a service disruption that affected the experience of receiving care without undermining therapeutic outcomes. The access barrier data confirms this: service wait time spiked from below 6% to 21.1% in 2023 and has since recovered. The 2023 dip was caused by a specific, identifiable operational problem that has been resolved.

Visit Frequency as the dominant predictor of satisfaction carries two possible interpretations: clients who visit more often become more satisfied over time through accumulated benefit, or satisfied clients are simply more likely to return. Survey data cannot distinguish between these. However, both interpretations point toward the same operational implication: retaining clients past the first visit is the highest-leverage action available. Whether increased frequency causes satisfaction or reflects it, the two are inseparable in practice.

The finding that Cancer Role does not significantly predict satisfaction is practically useful. It means the Center does not need differentiated programs targeting specific patient groups; the quality of service delivery appears consistent across current patients, survivors, and care partners. The strategic focus should be on engagement depth, not demographic targeting.

## 5.2.How the Results Can Be Used

The most immediately actionable findings come from three parts of the analysis. First, the access barrier trend shows that transportation barriers have grown in parallel with the aging of the client base. The Center can use this to justify investment in transportation support or scheduling changes before the problem becomes more severe. Second, the clustering results identify Cluster 2 as a specific, bounded target for intervention. Rather than trying to improve satisfaction across the board, the Center can focus resources on the 54 clients who are getting the least from their visits. Third, the frequency-satisfaction relationship provides a quantitative basis for prioritizing first-visit retention programs over other satisfaction initiatives.

# 6. Limitations

## 6.1. Issues Encountered

The most significant technical challenge was the non-uniform survey schema across years. Every year, the survey added new questions and repositioned or reworded existing ones. The satisfaction question alone moved from column 0 in 2022-2024 to column 4 in 2025 due to four new metadata columns. Service usage columns had entirely different positions in each year. Constructing and verifying the column index mappings required manual cross-referencing against actual Excel column names and sample values. A single incorrect index would silently produce wrong results without any error message.

A related issue is that PCA and K-Means were initially built using 2025 data only. During the review process, we identified that 12 equivalent service columns exist consistently across all four years. The feature matrix was rebuilt using all 1,179 rows. The earlier 2025-only models were based on approximately 292 rows, which substantially limited their reliability.

The Decision Tree classifier faces a fundamental class imbalance problem. With 89.2% of respondents reporting high satisfaction, the model is trained predominantly on one class. Even with class_weight='balanced', recall for the low-satisfaction class was only 0.43, meaning the model misclassified more than half of actual low-satisfaction clients. This limits the reliability of individual-level predictions.

## 6.2. Missing Data

The most consequential structural limitation is self-selection bias. The survey is administered only to clients who have already visited the Center and chosen to complete the questionnaire. Clients who visited once and never returned, likely the most disengaged segment, are systematically absent. All satisfaction and NPS figures therefore overestimate the experience of the full population the Center has ever served.

Service usage is encoded as binary: used or not used. The number of sessions attended is not captured. A client who received one massage and a client who received twelve sessions are represented identically. This prevents any analysis of dose-response relationships.

No unique client identifier links respondents across years. It is not possible to determine whether the same person completed the survey in multiple years, whether individual clients improved over time, or at what point clients typically disengage. All cross-year comparisons in this analysis describe population-level trends, not individual trajectories.

Racial and ethnic diversity is insufficient for subgroup analysis. With approximately 91%

of respondents identifying as White or Caucasian, no other racial or ethnic group has enough representation in any single year to support statistically valid comparisons.

# 7. Future Work and Recommendations

The following recommendations are drawn directly from the analysis findings. They are ordered by expected impact and implementation feasibility.

## 7.1. Design a Structured First Visit Follow-Up Program or Survey

Visit Frequency is the strongest predictor of satisfaction and the only variable confirmed as significant by both machine learning and statistical testing. The proportion of one-time visitors increased from 7.4% in 2022 to 9.0% in 2025. A targeted outreach to clients after their first visit, asking how it went and suggesting a specific next appointment, directly addresses the retention gap. This requires no additional survey infrastructure and can be piloted with minimal cost.

## 7.2. Conduct a Targeted Survey for Cluster 2 Clients

The 54 clients in Cluster 2 report a 4% Life Improvement Rate and average satisfaction of 2.81. A short follow-up survey specifically designed for clients who visited fewer than three times and reported low satisfaction would provide the Center with direct insight into why this group is not engaging. This analysis cannot answer that question from the existing data; a targeted instrument is required. This is a higher-value investment than any general satisfaction survey.

## 7.3. Address Transportation Barriers Before the Problem Compounds

Transportation barriers grew from 3.0% in 2022 to a peak of 13.3% in 2024, in parallel with the doubling of the 75-plus client population. If the demographic trend continues, transportation will become an increasingly significant obstacle. Transportation partnerships, subsidized ride programs, and expanded virtual service availability should be evaluated now rather than reactively.

## 7.4. Standardize the Survey Schema from 2026 Onward

Column positions, question wording, and response options changed in every survey year, making cross-year analysis significantly more difficult than it needed to be. Locking the survey structure from 2026 onward would allow the analysis pipeline in this project to run automatically on new data with minimal modification. New questions can be added to the end of the survey; existing questions should not be repositioned or reworded without maintaining backward compatibility.

## 7.5. Add a Unique Client Identifier

Adding a persistent identifier that links the same client across multiple years would transform this from a population-level study into a longitudinal one. The most valuable questions about the Center's impact, whether clients improve over time, at what point they disengage, and whether specific services produce durable outcomes, cannot be answered without individual tracking.

## 7.6. Apply Natural Language Processing to Open-Ended Responses

Every survey year includes three to four open-ended text fields with response rates between 30 and 50 percent. These were not analyzed in this project. Sentiment analysis and topic modeling on these fields would surface specific complaints, unmet needs, and positive experiences that are not visible in the quantitative data. This represents the highest-value untapped data source in the existing surveys.